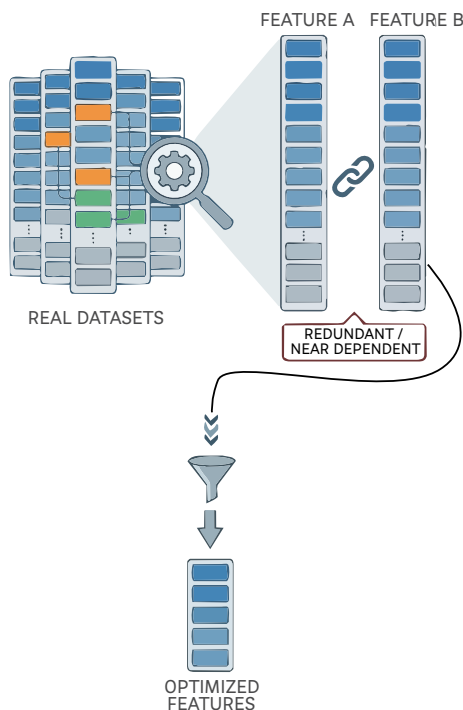


Detecting Redundant and Near-Dependent Features in Real Datasets

QR decomposition, a classical linear algebra technique, can be used to detect and eliminate linearly dependent and near-dependent columns in the feature matrix.



In machine learning, real-world datasets often suffer from multicollinearity—a phenomenon where feature vectors are linearly dependent or nearly dependent. This redundancy can lead to several critical issues:

- **Numerical instability:** Ill-conditioned matrices make matrix inversion unreliable.
- **High variance:** Unstable regression coefficients that are highly sensitive to small changes in data.
- **Overfitting:** Reduced model generalisation due to noise-sensitive features.

While traditional methods like correlation matrices identify pairwise relationships, they often fail to detect multivariate dependence. This article proposes a systematic approach using QR decomposition, a classical linear algebra technique, to detect and eliminate linearly dependent and near-dependent columns in the feature matrix.

Mathematical foundation

Let...

$$X \in \mathbb{R}^{m \times n}$$

...be a feature matrix, where:

- m = number of samples
- n = number of features

We now introduce QR decomposition as a principled alternative. For a given matrix X , it can be factored as

$$X = QR$$

...where:

- Q is an $m \times m$ orthogonal matrix, and
- R is an $m \times n$ upper triangular matrix.

In an upper triangular matrix, the diagonal elements reveal the linear dependencies of columns. If any of the diagonal elements are 0 or near 0, it indicates dependency.

Practical implementation in Python

Case A: Full rank matrix

In the case of a full rank matrix all the diagonal entries of R will be non-zero. Let's take an example:

```
import numpy as np

# Full rank 3x3 matrix
X_full = np.array([
    [1, 2, 3],
    [0, 1, 4],
    [5, 6, 0]
])

Q_full, R_full = np.linalg.qr(X_full)

print("Full Rank Matrix R:")
print(R_full)
print("Rank:", np.linalg.matrix_rank(X_full))
```